

Reinventing Igisoro: Exploring Materiality in a Cultural Heritage Game Board

Daniella Kalinda
Synaesthetic Media Lab
Toronto Metropolitan University
Toronto, Ontario, Canada
dkalinda@torontomu.ca

Marieme Lo
University of Toronto
Toronto, Ontario, Canada
marieme.lo@utoronto.ca

Ali Mazalek
Synaesthetic Media Lab
Toronto Metropolitan University
Toronto, Ontario, Canada
mazalek@torontomu.ca

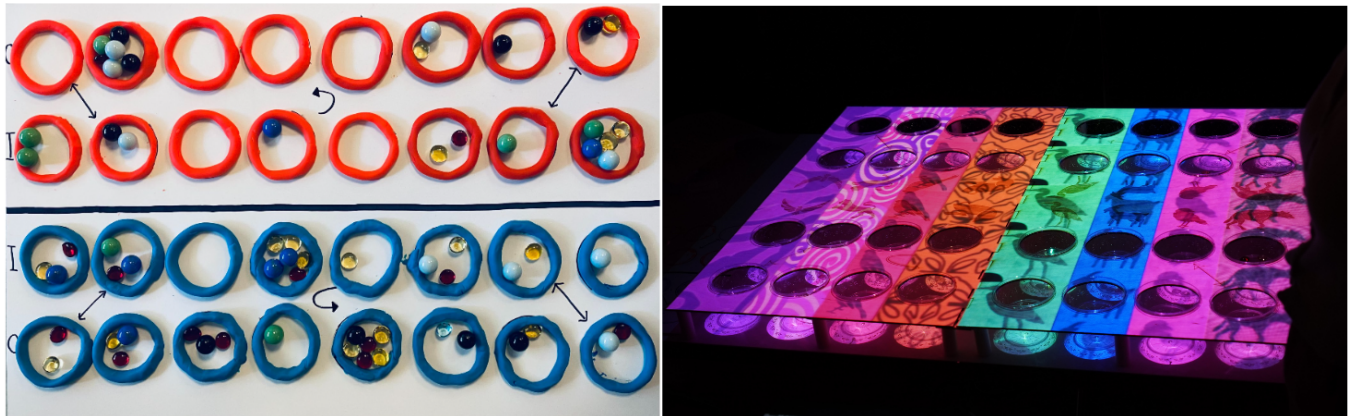


Figure 1: Two iterations of the igisoro board. Left: a Bristol board with clay ridges. Right: a wooden board with projections.

Abstract

We explore the material aspects of a traditional board game, Igisoro, to support connection to cultural heritage. We present two prototypes: the first prototype explores each material involved in the tactile interaction of the board. We conducted an informal design workshop, gathered feedback from our media lab, and applied relevant insights. The second prototype is computer-designed to enable user-friendly play and an upcoming narrative implementation. This prototype is enabled by Adobe Illustrator software, a laser cutting machine and a 3D-printer. We contribute a documentation of our material design process that provides preliminary design considerations for a tangible cultural heritage game that facilitates connections to intangible cultural heritage.

CCS Concepts

• **Human-centered computing** → *Interface design prototyping*.

Keywords

tangible interaction, materiality, cultural heritage

ACM Reference Format:

Daniella Kalinda, Marieme Lo, and Ali Mazalek. 2025. Reinventing Igisoro: Exploring Materiality in a Cultural Heritage Game Board. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3749275>

1 Introduction

As the world becomes more globalized, designing experiences around cultural connection is gaining momentum. Museums, especially, have taken this lead and incorporate innovative technologies in their exhibitions that are meaningful and relevant for visitors [3] [9]. The ‘new museum’ has shifted focus from conserving artifacts to authentic visitor experiences [1]. This interaction design of cultural heritage has inspired computer-supported research that is more relevant to today’s user. Outside the museum, the concept of family regroupement can inspire creative strategies for connection in the present [12]. That is, people in the diaspora find ways to keep their culture alive by adapting its materials in their new environment. The materials of a culture become catalysts for creativity to “reconstitute home, kinship and conjugality” [12]. Our work implicitly includes this sentiment in the material, cultural and play exploration.

Cultural heritage is an umbrella term for beliefs and practices throughout history that contribute to the knowledge and identity of an individual [3]. Exploring cultural heritage through tangible interaction is preferred by scholars [16] and play-based interpretations of cultural heritage prompt conversational memories that are otherwise fraught [14]. Tangible augmented reality experiences are considered more natural and are positively received by visitors [11],

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSCW Companion '25, Bergen, Norway
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3749275>

allowing visitors to engage with historical artifacts (or replicas) [8]. A tangible interface may give visitors an interactive surface to order a museum visit, such as a dynamic Hercules interface that represent heritage topics. The knowledge embedded in each piece augments the experience and shows the possibilities for digital output based on tangible input [20]. Other researchers have embedded recording devices in jewelry to intuitively 'carry' an oral storytelling practice captured in a beaded necklace [19]. Still others innovated a memory board where visitors created their own stories by combining tangible shells to digital beads [2]. Examples such as these pave a way for the meaningful design of tangible and embodied cultural heritage experiences.

Tactile interactions during storytelling give users a deeper attachment to the emotional connection of a cultural heritage piece [4]. Diegetic objects help users engage with a story-world on a physical platform [6]. It is likely that 3D printed replicas are useful for multi-modal learning [10]. Such tangible objects are intuitively used by visitors as interactive surfaces without prompting [15]. Also, gamifying learning experiences have been found to be the most rewarding for users, leading to a greater sense of ownership [17]. Designers may also choose to digitally augment a narrative since engaging with tangible historical artifacts was found to increase interest, engagement and emotional connection to the item [18]. Therefore, our research-through-design project explores how tangible cultural heritage, a game board traditionally passed down between generations, can serve as an interface for learning intangible cultural heritage. Drawing from the background of one of our team members, we worked with the traditional Rwandan board game, Igisoro. The game is simple yet unpredictable, giving plenty of opportunity for dynamic storytelling.

2 Igisoro

Igisoro is a 2-player board game of the mancala family of games. The board consists of 4 rows and 8 columns and begins with 4 seeds in each pit of the 2 innermost rows. The objective of the game is to capture as many of one's opponent's seeds as possible. A move consists of gathering seeds from a pit that contains at least more than one seed and 'sowing' one seed at a time in an anticlockwise direction until there is one seed in one pit. If the last seed lands in a pit that has another seed, the player must pick up all the seeds in that pit and continue to sow. Either player can start the game.

A capture can be made when a player lands a seed in the inner row and the opponent has seeds in both pits of the same column. The player then captures all their opponent's seeds within that column, and continues 'sowing' seeds in their area. The game continues for as long as a play can be made by both players. A player wins by capturing enough of the other player's seeds that no winnable moves are possible. For a more detailed explanation of the game and for more complex moves see [7].

3 Context and Goals

This work is a part of a larger research project on the role of immersive environments in enhancing youth engagement in cultural heritage. The project was conducted as part of a multidisciplinary research-based course on tangible media with graduate and undergraduate students. We investigated the role of materiality and

the interaction design of intangible cultural heritage during gameplay. To do so, we picked Igisoro, a traditional Rwandan game, as a tangible and embodied interface to evoke cultural heritage. We took an intentionally collaborative approach to working together and incorporating each person's unique skills and perspectives to create moments of learning about heritage. In our design, we explored the felt experience by exploring the material interaction at the board level, the pit level, the seed level, and the thematic embodied experience at the intangible heritage level. For intangible heritage, we conducted an informal workshop with users to find possible points of connection for cultural meaning. Our research question in this design exploration: In what ways does materiality inform the design considerations for embedding computer-enabled intangible cultural heritage during a gameplay experience? Our research goals in this design exploration include (1) Exploring material interactions during each aspect of tangible gameplay to support connection to cultural heritage, and (2) Finding possible intangible cultural heritage points of interaction during gameplay.

4 Paper Prototype

Board: We sought to maintain the physical integrity of the game board and rules. The first paper prototype uses sticky notes to indicate where the pits would be placed (see Figure 3, top-left). This allowed us to materially familiarize ourselves with the board structure and the gameplay. Next, we moved to a Bristol board with clay circles as pits and tested the added visual cues: directional arrows and lettering (see Figure 1). The initials "I" and "O" represent the "inner" and "outer" areas of the board. Arrows pointing anticlockwise are at the center of the board to direct game flow and arrows for where clockwise turns are allowed (not used in this iteration to reduce complexity).

Seeds: We explored the shape, size and texture of game seeds including clay beads, authentic seeds, round marbles and flat pebbles (see Figure 2). We found regular clay did not harden after several weeks. Natural seeds were less than ideal for picking up and dropping the seeds at a fast pace. Yet, "good" service design practice teaches that design fades to the background and enables a user's intended task [13]. Based on this, we used the round marbles and flat pebbles to explore the gameplay experience in the workshop, which is described in the following section.



Figure 2: We explore different seed types including (from left to right) play dough, pumpkin seeds, marbles, and flat marbles.

Pits: There are 32 pits on an igisoro board. As mentioned above, we added clay borders as proto-pits and examined the pit's design and necessary elements. We also tested the use of plastic cups (not

pictured) and found that it had a positive effect on picking up and dropping seeds but would require a stronger board structure that would be implemented in a later iteration.

5 Workshop

The team used a university course on tangible media to conduct a design workshop in which peers interacted with the paper prototype and provided feedback in an informal discussion. There were 21 participants in the workshop. We tested how players learned to play the game through instruction and visual cues on the game board, the tangibility and ease of use of the seeds, and the integration of the intangible cultural heritage. We created three game boards where two contained visual cues indicating the "inner" and "outer" areas of the board and the remaining board was blank. We aimed to compare how players learned the game with and without the visual cues on the board.

We had three pairs of players test out the prototype at a time (see Figure 3, right). Roughly half of the workshop participants played the game, while the rest observed the gameplay. The players were presented with a Bristol board with circular clay rings that acted as pits and with a selection of different kinds of seeds. We instructed the players on the rules of the Igisoro game and assisted them when they were unsure about the game's rules. We observed how players interacted with the different seeds on the board, and how they picked them up and moved them. The seeds included different sized marbles, glass pebbles, and decorative stones. We introduced a folkloric story to workshop participants and described ways in which the story elements could be integrated into the gameplay. During gameplay, we discussed the best points of interaction to trigger a story projection. After gameplay, we engaged in a collaborative design discussion with workshop participants about the effectiveness of the tangible game interface and about the integration of intangible cultural heritage to the gameplay.



Figure 3: Paper prototype with sticky-notes and marbles (top-left), Original board with beads (bottom-left), Design workshop with Bristol board, clay pits and 'seeds' (right).

Lessons Learned:

- Game board: the game board needed deeper pits to facilitate gameplay.
- Seeds: Flat, medium sized glass pebbles worked best as game seeds. Large marbles proved to be too large and the pits were unable to hold numerous seeds at once.
- Visual cues: The 'I' and 'O' lettering on the side of the board was not very useful as a cue about the inner and outer parts

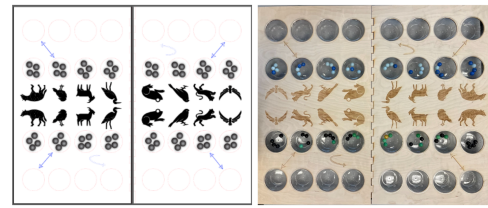


Figure 4: Computer design and implementation of board from 13x18 Baltic birch wood with engravings of totems.

of the board. However, the directional arrows proved to be useful in reminding players about the direction of the game and the places where they can make alternative moves. We observed that the arrows were more self-evident markers for the players than lettering.

- Intangible heritage integration: We found that incorporating a folkloric story during gameplay interrupted its natural flow instead of enriching the game experience. A less linear narrative was recommended. One suggestion was to share about the history of the kingdom/country of Rwanda that is more general. We incorporate this feedback in the second prototype.

Our design exploration and workshop helped identify areas for improvement for the tangible design of the board, pit and seeds. We were given design considerations for the interactive design of the intangible cultural heritage. Workshop participants found that the fable disconnected them from the game itself and would prefer a non-linear narrative around the history of Rwanda to allow for flexibility in the gameplay. The arrow cues on the gameboard were found to be helpful to new Igisoro players but the lettering was not needed. We explored the weight, texture, and color of the seeds in the design of our first prototype, finding that flat glass beads were preferred by game players.

6 Wooden Prototype

Based on the feedback from the design workshop, our approach to creating the second prototype of an augmented Igisoro cultural experience was based around the gameplay interactions coupled with a more organic and open-ended narrative. Gameplays on the board are intended to trigger historical clan representations projected over the board. Each column of the board represents a clan, and is shown by the engraved totem.

Board: The second prototype was designed on Adobe Illustrator (see Figure 4, left) and laser cut from a thin 30x18 Baltic Birch wood (see Figure 4, right). The scent of the wooden board added an organic connection to the gameplay, reminiscent of the original elements of Igisoro. We rastered the directional arrows as visual cues for the new player, as these cues were successful in our first prototype. We extended the space at the center of the board for narrative storytelling and rastered eight totems on the board's surface twice to face each player. The board is raised on 3D printed stilts to allow the pits to lay flush on the board.

Seeds: As noted above, the flat glass pebbles were chosen due to their shape, size, and texture. The seeds' semi-transparent material reflects light which is interesting for a projection implementation.

The color was also a consideration when choosing the seeds. The glass pebbles came in various blue hues that stood out in the clear pits.

Pits: Based on feedback from the design workshop, we chose to create pits that recessed into the board's surface so players can easily scoop up the seeds. The pit bowls are made from clear, small plastic bowls with a rim that lays flat on the board's surface. The bowls' clear material may enhance an augmentation iteration as light can pass through the bowls.

Historical Context: A clan refers to a group of people who share a common ancestor. The clans we chose represent the 8 most populated clans in the kingdom of Rwanda [5]. Sharing narrative information about clans through audio and visual projections provides the player with intangible cultural heritage to enrich the game experience. We used the totems of each clan as diegetic visual cues for intangible heritage (see Table 1 and Table 2). We integrated these totems in the design of the game board, where each totem was placed above a column of pits signifying a placeholder for storytelling.

Table 1: The first 4 major clans and totems.

Name	Abanyiginya	Abega	Abazigaba	Abagesera
Totem	Crane	Frog	Leopard	Wagtail

Table 2: The next 4 major clans and totems.

Name	Abasinga	Abacyaba	Ababanda	Abashambo
Totem	Eagle	Hyena	Crow	Cattle

7 Conclusion and Future Work

We explored the materiality of a cultural heritage game to inform computer-enabled design decisions in heritage storytelling. Our first prototype used a Bristol board with clay ridges for pits and tested various 'seeds' for game pieces. We added directional arrows as visual cues to help new users remember game mechanics. Yet, our design workshop demonstrated how new users interact with the game board. Discussions from the workshop revealed that a linear narrative constricts game play so we explored a general historic theme. Our second prototype was computer-enabled design; a laser cut and machine-rastered wooden prototype that more closely resembles the original board. We used medium-sized flat glass pebbles as seeds. Eight clan totems were rastered onto each column as diegetic visual cues. Future work involves more diegetic game pieces with interactive projections that respond to gameplay decisions and user evaluations.

Acknowledgments

We thank Cadance Tan, Anitha Nathan, Hayley Kingstone and Melony Manikavasgavar for their invaluable contribution and our peers for their insightful feedback.

References

- [1] Loris Barbieri, Fabio Bruno, and Maurizio Muzzupappa. 2018. User-centered design of a virtual reality exhibit for archaeological museums. *International Journal on Interactive Design and Manufacturing (IJIDeM)* 12, 2 (2018), 561–571.
- [2] Jean Ho Chu, Paul Clifton, Daniel Harley, Jordanne Pavao, and Ali Mazalek. 2015. Mapping Place: Supporting Cultural Learning through a Lukasa-inspired Tangible Tabletop Museum Exhibit. In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '15)*. ACM, New York, NY, USA, 261–268.
- [3] Areti Damala, Ian Ruthven, and Eva Hornecker. 2019. The MUSETECH model: A comprehensive evaluation framework for museum technology. *Journal on Computing and Cultural Heritage (JOCCH)* 12, 1 (2019), 1–22.
- [4] Daniel Echeverri and Huaxin Wei. 2021. Designing physical artifacts for tangible narratives: lessons learned from letters to José. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [5] SW Freedman, Harvey M Weinstein, and Timothy Longman. 2006. The teaching of history of Rwanda: A participatory approach. *Human Rights center, University of California: Berkeley* (2006).
- [6] Saumya Gupta, Theresa Jean Tanenbaum, and Karen Tanenbaum. 2019. Shiva's Rangoli: Tangible storytelling through diegetic interfaces in ambient environments. In *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, New York, NY, USA, 65–75.
- [7] Dirk Huylebrouck. 1996. Puzzles, Patterns Drums: The Dawn of Mathematics in Rwanda and Burundi. *Humanistic Mathematics Network Journal* 1, 14 (1996), 6.
- [8] Suzanne Kobeisse. 2021. Touching the Past: Developing and Evaluating a Heritage kit for Visualizing and Analyzing Historical Artefacts Using Tangible Augmented Reality. *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* 73 (2021), 1–3.
- [9] Markos Konstantakis and George. Caridakis. 2020. Adding culture to UX: UX research methodologies and applications in cultural heritage. *Journal on Computing and Cultural Heritage (JOCCH)* 13, 1 (2020), 1–17.
- [10] Jamie Kwan, Jean Ho Chu, Daniel Harley, Melanie McBride, and Ali Mazalek. 2016. Grasping cultural context through multisensory interactions. In *Proceedings of the TEI'16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, New York, NY, USA, 482–487.
- [11] Panayiotis Kyriakou and Sorin Hermon. 2019. Can I touch this? Using natural interaction in a museum augmented reality system. *Digital Applications in Archaeology and Cultural Heritage* 12 (2019), e00088.
- [12] Marieme Soda Lo. 2015. Senegalese immigrant families' 'regroupement' in France and the im/possibility of reconstituting family across multiple temporalities and spatialities. *Ethnic and Racial Studies* 38, 15 (2015), 2672–2687.
- [13] Downe Lou et al. 2020. 15 Principles of Good Service Design. *NASWA Workforce Technology* (2020).
- [14] Anders Sundnes Løvlie, Karin Ryding, Jocelyn Spence, Paulina Rajkowska, Annika Waern, Tim Wray, Steve Benford, William Preston, and Emily Clare-Thorn. 2021. Playing games with Tito: Designing hybrid museum experiences for critical play. *Journal on Computing and Cultural Heritage (JOCCH)* 14, 2 (2021), 1–26.
- [15] Mark T Marshall, Nick Dulake, Luigina Cioffi, Daniele Duranti, Hub Kockelkorn, and Daniela Petrelli. 2016. Using tangible smart replicas as controls for an interactive museum exhibition. In *Proceedings of the TEI'16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction*. 159–167.
- [16] Anitha Nathan, Daniella Kalinda, Lucas Hrynshyn, Ellen Reade, Ravit David, and Ali Mazalek. 2022. Designing a tangible augmented reality experience for cultural heritage research. *DRS2022: Bilbao* 25 (2022).
- [17] Eslam Nofal, Georgia Panagiotidou, Rabee M Reffat, Hendrik Hameeuw, Vanessa Boschloos, and Andrew Vande Moere. 2020. Situated tangible gamification of heritage for supporting collaborative learning of young museum visitors. *Journal on Computing and Cultural Heritage (JOCCH)* 13, 1 (2020), 1–24.
- [18] Elena Not, Dario Cavada, Stefano Maule, Anna Pisetti, and Adriano Venturini. 2019. Digital augmentation of historical objects through tangible interaction. *Journal on Computing and Cultural Heritage (JOCCH)* 12, 3 (2019), 1–19.
- [19] Lizette Reitsma, Andrew Smith, and Elise. van den Hoven. 2013. StoryBeads: Preserving indigenous knowledge through tangible interaction design. In *2013 International conference on culture and computing* (2013), 79–85.
- [20] Stéphanie Rey, Célia Picard, Yanis Fatmi, Fanny Franco, Sarah Guilbert, Jérémie Manéré, Christophe Bortolaso, Mustapha Derras, Nadine Couture, and Anke M. Brock. 2020. Build your own hercules: Helping visitors personalize their museum experience. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction* (2020), 495–502.